

Chapter 8

GIS/Mapping

In this chapter...

- **Learn about BenMAP-CE's mapping functions.**
- **Map different variables and modify the map display.**

The BenMAP-CE Geographic Information System (GIS) will display maps of air quality, health and economic data. These maps can help answer a number of questions:

- *Quality assurance*: Do your air quality changes seem to be distributed correctly? Are your air quality changes and health impacts occurring in approximately the same location?
- *Presentations*: In what states or provinces are most of the benefits/disbenefits of your policy scenario concentrated?
- *Analysis*: Which air quality grid cells contain the highest ozone values?

The main **GIS Map** tool will be available once you have successfully completed the first stage of the BenMAP-CE analysis (**Air Quality Surfaces** in the main BenMAP-CE tree menu).

8.1 Overview of Mapping

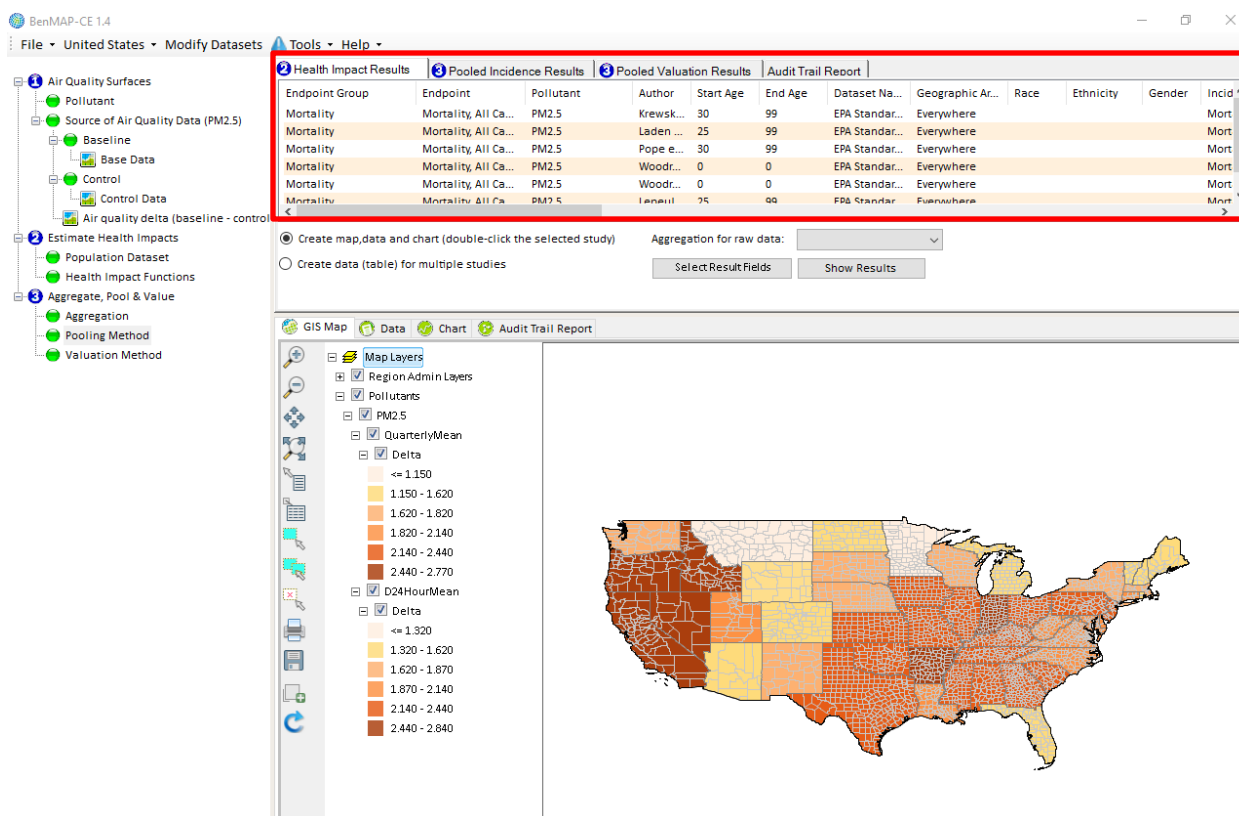
The GIS will map three categories of data:

1. *Air quality (.aqgx)*. Air quality grid maps represent summary air quality metrics (e.g., daily average, daily maximum, or other metric where available) within each grid cell. Air Quality Grids can be added to the **GIS Map** by following the steps in Chapter 5: Creating Air Quality Grids.
2. *Incidence (.cfgrx)*. A configuration results file contains the results of your analysis reported at each air quality grid cell. These results have not been aggregated, pooled or valued. For more information on choosing configuration settings, see Chapter 6: Estimating Incidence.
3. *Aggregated, pooled and valued results (.apvrx)*. These are results that have been aggregated to a coarser spatial scale, see Chapter 7: Aggregation, Pooling, and Valuation.

When mapping APV Configuration results, you can generate six different types of maps: (1) *Incidence*, (2) *Aggregated Incidence*, (3) *Pooled Incidence*, (4) *Valuation*, (5) *Aggregated Valuation*, and (6) *Pooled Valuation*. For more information on how to load these files into BenMAP-CE, see Chapters 6 and 7 (Estimating Incidence, and Aggregation, Pooling, and Valuation).

8.2 Results Panel

Once you have gone through all the steps outlined in the previous chapters to import data and files, you are ready to begin exploring the visual results. Each of your selected endpoint groups will be displayed in the results panel (upper right portion of the main BenMAP-CE window).

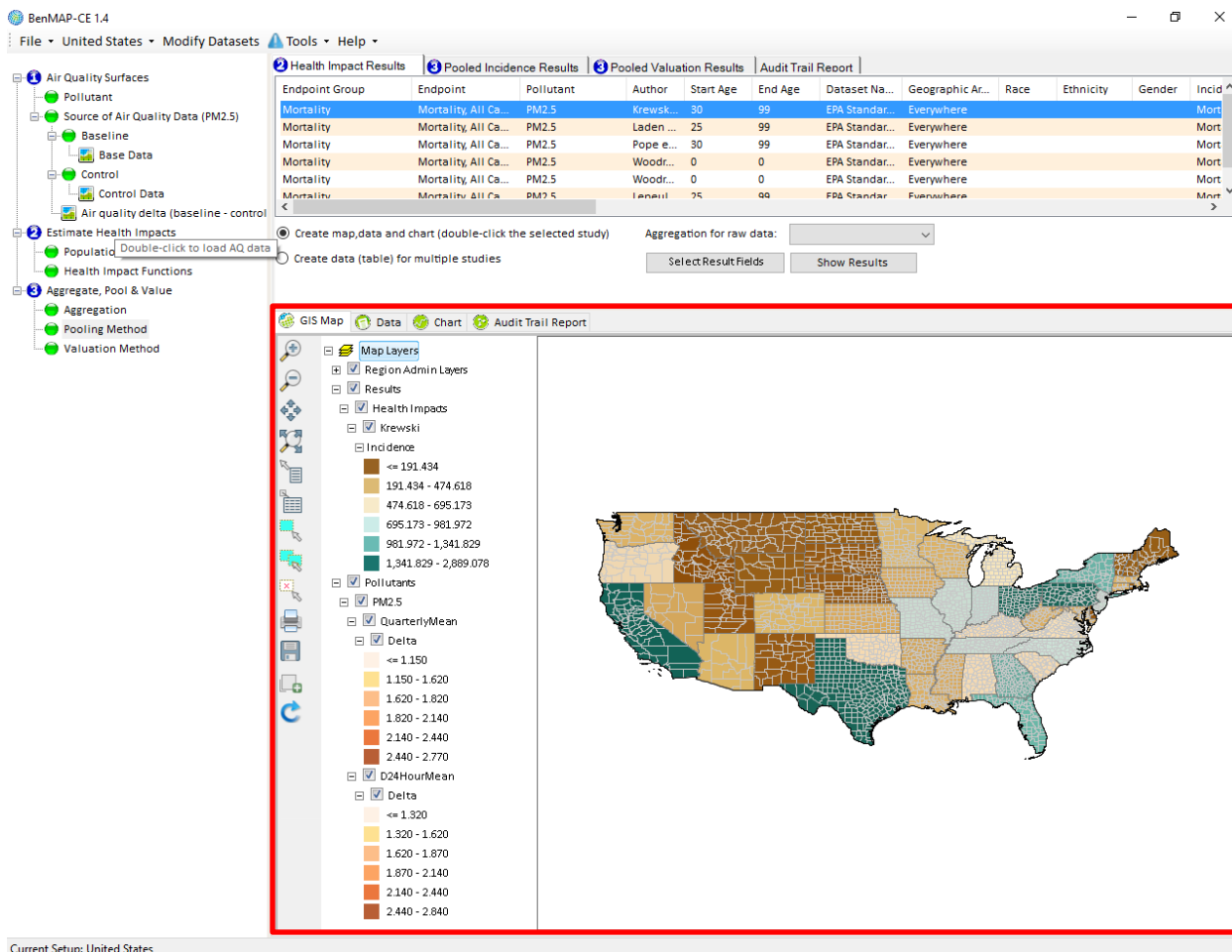


There are four tabs within the results panel, **Health Impact Results**, **Pooled Incidence Results**, **Pooled Valuation Results**, and **Audit Trail Report**. To create a results map layer for an endpoint group, select an entry, or entries, from the list and drag it down onto the GIS Mapping panel. BenMAP-CE will create the map and place an entry into the GIS table of contents below the newly created results group. The layer will most likely be listed under the author's name as subgroup of the main results group. This process of dragging-and-dropping can be done with as many entries as you like, on any of the first three tabs.

Audit Trail Reports facilitate transparency and reproducibility by reporting a summary of your assumptions underlying each step of the analysis. This is described in more detail below.

8.3 GIS Mapping Panel

The GIS Mapping panel is the centerpiece of this tool. Here is where you can view, edit, add, and remove layers from the table of contents and GIS map viewer.



8.3.1 GIS Map Tab

After creating an air quality surface, the **GIS Map** tab is used to view the air quality data (double-click on an available air quality surface to display it). Here you will find a GIS table of contents, toolbar, and interactive map.

8.3.1.1 GIS Table of Contents

The GIS table of contents is where you will find all your loaded map layers. The layers are sorted in groups, with subgroups below them. The general setup will include *Region Admin Layers* group, *Pollutants* group, and *Results* group. You can select or deselect as many layers or groups as you like, for viewing on the map to the right.